Assignment 1

Due: 16.10.2026

1. Introduction

In this assignment, you will use basic R programming to simulate how insecticide resistance evolves in an insect population.

Consider a single gene with two alleles: `S` (susceptible to the insecticide, dominant) and `r` (resistant, recessive). Because `r` is recessive, only insects with genotype `rr` clearly survive spraying, insects with genotype `SS` or `Sr` still behave as susceptible. We describe this with a survival probability between 0 and 1 for each genotype, for example:

```
w_SS <- 0.35   # the probability of surviving to reproduce of genotype SS
w_Sr <- 0.35   # the probability of surviving to reproduce of genotype Sr
w_rr <- 1.0    # the probability of surviving to reproduce of genotype rr
```

Let q_n denote the relative frequency of the resistance allele `r` in generation n, and let $p_n = 1 - q_n$ be the relative frequency of `S`. Assuming random mating, the genotypes `SS`, `Sr` and `rr` occur in generation n with the Hardy-Weinberg proportions p_n^2, $2p_nq_n$ and q_n^2. After one round of selection by the insecticide, the frequency of the resistance allele in generation $n+1$ is

$$q_{n+1} = \frac{p_n q_n \, w_{Sr} + q_n^2 \, w_{rr}}{\bar{w}_n}, \qquad \bar{w}_n = p_n^2 \, w_{SS} + 2p_n q_n \, w_{Sr} + q_n^2 \, w_{rr},$$

where $\bar{w}_n$ is the mean survival probability in generation n.

2. Task

Create a new R script called `Assignment1_solution.R`.

1. Create a variable for the number of generations to simulate (`25`).
2. Create variables `w_SS`, `w_Sr` and `w_rr` for the three genotype survival probabilities given above.
3. Create a vector `generation` containing the numbers 0 to `number_of_generations`.
4. Create a numeric vector `q` of length `number_of_generations + 1`, and set `q[1] <- 0.05` as the starting frequency of the resistance allele, before spraying began.
5. Use a `for` loop to fill in the remaining entries of `q` based on the formulas above.
6. Modify the loop to also save the mean survival probability $\bar{w}_n$ in a vector `w_bar` of length `number_of_generations`.
7. To see how the strength of the insecticide affects the speed of resistance evolution, repeat the simulation for a weaker insecticide (`w_SS <- 0.6` and `w_Sr <- 0.6`). Use a `for` loop, as before, to compute this second frequency trajectory and store it in a new vector `q_weak`.

8. To see how the starting frequency of the resistance allele interacts with the strength of selection, repeat the simulation once more using a lower starting frequency `q[1] <- 0.01` combined with a much stronger insecticide (`w_SS <- 0.1` and `w_Sr <- 0.1`). Use a `for` loop, as before, to compute this trajectory and store it in a new vector `q_lowstart`. Even though it starts out behind the other two scenarios, the stronger selection should let it overtake both of them at some point during the simulation.

9. Use a `while` loop to find the first generation in which `q_lowstart` overtakes `q`, and print that generation.

10. Create a plot of `q`, `q_weak` and `q_lowstart` against `generation`. Add a title, labels for both axes and a legend distinguishing the three scenarios.

11. Add a short comment explaining what the simulation suggests about the effect of insecticide strength and starting frequency on how quickly resistance spreads.